

WeiJia Zhang

2172003915 | zhangwj.charlie@gmail.com | [linkedin.com/in/weijia-charlie-zhang/](https://www.linkedin.com/in/weijia-charlie-zhang/) | [charliedreemur.github.io](https://github.com/charliedreemur) | [Google Scholar](#)

EDUCATION

Yale University <i>M.S. in Computer Science (Thesis Track with Full Scholarship)</i>	Sep 2026 - May 2028 <i>GPA: 4.0/4.0</i>
University of Illinois Urbana-Champaign (UIUC) <i>B.S in Computer Science and Mathematics C.W. Gear Outstanding Undergraduate Student (One of two annual recipients)</i>	Aug 2022 - May 2026 <i>GPA:3.7/4.0 (Dean's List)</i>

WORK EXPERIENCE

TikTok <i>Research Intern</i>	Jun. 2026 – Present
- Designed a self-evolving content-moderation agent framework centered on a ReAct agent that autonomously analyzes individual videos using specialized tools for frame sampling, ASR/OCR, metadata extraction, originality scoring, and watermark/celebrity detection; integrates multimodal evidence with policy-grounded reasoning to produce structured moderation decisions. - Built a self-evolution loop where a <i>Reflector</i> aggregates Solver failures into root-cause attribution reports, and an <i>Evolver</i> add/revise/merge case-relevant skills to refine the Solver and improve 15.5% overall f1 score	
Microsoft <i>Research Intern</i>	July 2025 – Sep 2025
- Conducted LLM research for Microsoft 365 Excel Copilot, focusing on spreadsheet text-analysis And Office.js code generation. - Developed the TextAnalysisSFT data pipeline for generating SFT data for Excel Copilot’s TextAnalysis API: mined 2,000+ real-world examples from Kaggle, filtered text-heavy spreadsheets, generated user queries and Office.js programs, and validated executions with SheetEngine; the resulting dataset improved Office Scripts code-generation accuracy by 15%.	
Reborn Network <i>AI Engineer</i>	May 2023 – July 2023
- Developed a role-playing agent Unity VR Game enabling agents interaction through text/voice/VR actions in real time (< 1s latency) - Introduced RAG/vector database to strengthen long-term agent memory, improving dialogue coherence score from 2/5 to 4/5 - Designed a reusable ”character card” framework, enabling UGC ecosystem and reducing character persona configuration time by 300%	
Tencent <i>Software Engineer, Wechat Group</i>	Aug 2024 – Sep 2024
- Developed a cross-platform (Android & iOS) hardware performance analysis tool for WeChat Mini Programs, supporting 200+ partner teams in identifying performance bottlenecks - Built Unity Mono Memory Profiler that discovered 40+ hidden memory allocation points, reducing memory-leak-related crash rate by 120%	

PUBLICATIONS(100+ CITATION)

WeiJia Zhang et al. “CUADebug: Diagnosing and Repairing Computer-Use Agent Failures” <i>EMNLP 2026</i> .
WeiJia Zhang et al. “Every Act Has Its Price: Compressed Moral Composition in Frontier LLMs.” <i>EMNLP 2026</i> .
WeiJia Zhang et al. “How Much Vision Does Multimodal Reasoning Need?” <i>Under review at NeurIPS 2026</i> .
WeiJia Zhang* , Zijia Liu*, Haoru Li*, et al. “SeeingEye: Agentic Information Flow Unlocks Multimodal Reasoning in Text-only LLMs.” <i>arXiv:2510.25092; under review at AAAI 2026</i> .
Miaomiao Li et al. (including WeiJia Zhang). “Understanding and Mitigating the Bias Inheritance in LLM-based Data Augmentation on Downstream Tasks.” <i>ACL 2026. Oral</i> .
Gordon Dai* and WeiJia Zhang* et al. “Artificial Leviathan: Exploring Social Evolution of LLM Agents Through the Lens of Hobbesian Social Contract Theory.” <i>AI Behavioral Science Workshop, ACM SIGKDD 2024</i> .
Kunlun Zhu et al. (including WeiJia Zhang). “Where LLM Agents Fail and How They Can Learn From Failures.” <i>arXiv:2509.25370. Failure Modes in Agentic AI, ICML 2026</i>

PROJECT EXPERIENCE

OpenManus & OpenManus-RL (Researcher, GitHub 60k+ Stars) GitHub	Jan 2025 – Jul 2025
- Collaborated with the official OpenManus team as a core contributor to OpenManus-RL, the follow-up project of OpenManus; the open-source ecosystem has accumulated 60,000+ GitHub stars - Implemented a ReAct-style reasoning architecture (<i>Think</i> → <i>Act</i> → <i>Tool-use</i>) and multi-agent collaboration workflows; built an MCP-based tool-library framework for automated tool invocation and output processing - Constructed high-quality SFT datasets for agent benchmarks including AlfWorld and WebShop; designed step-level reward signals to improve agent planning and memory capabilities	
CUAAGentDebugger (First Author / Lead)	Jan 2026 – Present
- Designed a GUI-agent error taxonomy with 4 major categories and 29 subtypes, covering perception, interaction localization, task reasoning, and external system failures across mainstream OSWorld/CUA scenarios - Built a self-evolving VLM-agent debugging framework that automatically identifies root causes from failed trajectories, distills them into reusable debugging skills, and enables agents to learn autonomously from historical failures - Designed a dual-layer memory architecture with episodic and semantic memory, together with an intent-aware RAG retrieval mechanism that injects skills from similar-intent trajectories to improve diagnosis and re-rollout accuracy	
Understanding and Mitigating Bias Inheritance in LLM-based Data Augmentation	Nov 2024 – Feb 2025
<i>Researcher</i>	
- Executed 50+ fine-tuning experiments using Llama-3.1 and ChatGPT-o1-mini with controlled bias blend ratios - Developed automated pipeline for synthetic data generation, bias injection, LoRA fine-tuning, and evaluation	

SKILLS

Programming Languages: Python, C++, C#
Frameworks & Libraries: VERL, VLLM, LangGraph, LangChain, PyTorch, Tensorflow
AI Focus: SFT, RL/Post-training, LLM Agents, Multimodal Reasoning, RAG, Self-Evolving Agent, MAS,